

Thesis Proposal: Functional Forecasting of Turkish Electricity Load Curves

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Research Question

Main question

Can a functional time-series model improve day-ahead forecasts of Turkish hourly electricity demand relative to the official forecast?

- The forecasting object is the full 24-hour daily load curve, not only a daily aggregate.
- Functional representation preserves intraday structure: overnight load, morning ramp, daytime peak, and evening decline.
- Forecasts are evaluated against two reference points:
 - seasonal naive benchmark, Y_{d-7} ;
 - official day-ahead forecast, $\hat{Y}_d^{off}$.

Empirical Object

Let $Y_d(h)$ denote Turkish electricity load on day d , hour $h = 0, \dots, 23$.

$$Y_d = \{Y_d(h) : h = 0, \dots, 23\}.$$

The cleaned daily matrices are

$$Y = [Y_d(h)]_{d,h}, \quad \hat{Y}^{off} = [\hat{Y}_d^{off}(h)]_{d,h}, \quad E = [Y_d(h) - \hat{Y}_d^{off}(h)]_{d,h}.$$

- Each retained day has 24 hourly observations.
- Evaluation starts on 2023-01-01.
- The evaluation sample contains 26,304 hourly observations.

Functional Representation

Each curve is first smoothed with a B-spline basis:

$$Y_d(h) \simeq \sum_{m=1}^M c_{dm} B_m(h), \quad M = 12.$$

FPCA then represents daily load curves as

$$Y_d(h) = \mu(h) + \sum_{k=1}^K \xi_{dk} \phi_k(h) + e_d(h).$$

The corresponding forecast is

$$\hat{Y}_d(h) = \hat{\mu}(h) + \hat{\Phi}(h)' \hat{\xi}_d, \quad \xi_d = (\xi_{d1}, \dots, \xi_{dK})'.$$

Choice of Functional Dimension

The baseline keeps $K = 2$ principal components.

FPC	Load var.	Load cum.	Error var.	Error cum.
1	93.40%	93.40%	76.60%	76.60%
2	3.89%	97.29%	12.00%	88.60%
3	1.11%	98.40%	4.62%	93.22%
4	0.99%	99.39%	2.88%	96.09%

- The first two load components explain 97.29% of total functional variation.
- $K = 2$ exceeds the usual 95% cumulative-variance rule.
- The first component captures daily level; the second captures the dominant intraday shape contrast.

Model-Building Strategy

The proposed model sequence is intentionally incremental.

- ① Reduce each daily load curve to a small vector of FPCA scores.
- ② Forecast scores using weekly and one-day lagged functional dynamics.
- ③ Add predetermined calendar and ramp variables.
- ④ Add holiday and lagged load-shape variables.
- ⑤ Add derivative and targeted intraday transition features.
- ⑥ Treat residual correction as a possible extension, not the baseline claim.

Forecasting Specifications

Benchmarks:

$$\hat{Y}_d^{SN}(h) = Y_{d-7}(h), \quad \hat{Y}_d^{OFF}(h).$$

Weekly FPCA VAR:

$$\xi_d = a + A_7 \xi_{d-7} + u_d.$$

FPCA VAR(1,7):

$$\xi_d = a + A_1 \xi_{d-1} + A_7 \xi_{d-7} + u_d.$$

Augmented score model:

$$\xi_d = a + A_1 \xi_{d-1} + A_7 \xi_{d-7} + Bz_d + u_d,$$

where z_d is predetermined at the forecast date.

Predictor Blocks

The largest predictor vector is

$$z_d = \left(z_d^{cal'}, z_d^{hol'}, z_d^{shape'}, z_d^{deriv'} \right)'.$$

- Calendar controls: weekend indicator and annual seasonality terms.
- Holiday controls: fixed holidays, religious holidays, holiday eves, post-holiday days, near-holiday days, and bridge days.
- Lagged load-shape variables: previous-day mean, peak, minimum, range, and rolling means.
- Ramp variables: morning, evening, and end-of-day load changes.
- Derivative summaries: mean slope, slope volatility, maximum slope, minimum slope, and slope range.

Evaluation Design

All forecasts use a rolling-origin design: for target day d , estimation uses only information dated $< d$.

Forecast errors:

$$e_{dh}^{(m)} = Y_d(h) - \hat{Y}_d^{(m)}(h).$$

Accuracy metrics:

$$MAE_m = \frac{1}{N} \sum_{d,h} |e_{dh}^{(m)}|, \quad RMSE_m = \left[\frac{1}{N} \sum_{d,h} (e_{dh}^{(m)})^2 \right]^{1/2}.$$

Gain relative to the official forecast:

$$Gain_m^{MAE} = 100 \frac{MAE_{OFF} - MAE_m}{MAE_{OFF}}.$$

Preliminary Results

Model	MAE	RMSE	MAE gain	RMSE gain
Seasonal naive Y_{d-7}	1912.46	3129.98	-57.60%	-79.85%
Official forecast	1213.50	1740.30	0.00%	0.00%
FPCA VAR(1,7) scores	1442.61	2018.30	-18.88%	-15.97%
FPCA VAR(1,7) + ramp/calendar	1238.37	1716.30	-2.05%	1.38%
FPCA VAR(1,7) + holiday/load-shape	1000.62	1353.56	17.54%	22.22%
FPCA VAR(1,7) + holiday/load-shape/derivative	980.38	1324.40	19.21%	23.90%

- The baseline proposal model improves on the official forecast by 19.21% in MAE.
- It improves on the official forecast by 23.90% in RMSE.
- Relative to the weekly seasonal naive benchmark, the MAE reduction is 48.74%.

Parsimony and Robustness

Specification	K	Resid.	MAE	RMSE	MAE gain	RMSE gain
Holiday/load/derivative	2	No	980.38	1324.40	19.21%	23.90%
Holiday/load/derivative	2	Yes	652.90	982.60	46.20%	43.54%
Holiday/load/derivative	4	No	800.99	1109.69	33.99%	36.24%
Holiday/load/derivative	4	Yes	640.10	967.82	47.25%	44.39%

- $K = 2$ is the conservative baseline because it is parsimonious and variance-rule motivated.
- $K = 4$ retains additional intraday shape variation and improves no-residual accuracy.
- Residual correction is promising empirically, but requires further methodological justification.

Residual Correction as an Extension

For target day d , fitted residual curves are computed only for days strictly before the target:

$$\hat{e}_s(h) = Y_s(h) - \hat{Y}_s^{base}(h), \quad s < d.$$

A correction curve is estimated from past residuals:

$$\hat{Y}_d^{res}(h) = \hat{Y}_d^{base}(h) + \lambda_d \hat{r}_d(h).$$

- The residual regression uses only information dated before d .
- The validation choice of λ_d also uses only pre- d information.
- This preserves the rolling-origin structure, but the extension is kept separate from the main thesis claim.

Expected Thesis Contribution

- Recast Turkish day-ahead electricity load forecasting as a functional forecasting problem.
- Compare functional forecasts directly with the official day-ahead forecast.
- Quantify the contribution of calendar, holiday, load-shape, ramp, and derivative information.
- Evaluate forecasts under a leakage-aware rolling-origin design.
- Clarify the tradeoff between parsimonious FPCA score models and richer shape-retaining alternatives.

Next Steps

- ① Formalize the final baseline specification and robustness checks.
- ② Decide whether $K = 4$ should remain a robustness check or become part of the main model class.
- ③ Strengthen the methodological justification for residual post-processing.
- ④ Add hour-level diagnostics to show where the functional model improves most.
- ⑤ Connect forecast gains to economic intuition about intraday demand and system operation.

Selected References

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